1 Lecture 18: Plan for Class

We are skipping the rest of chapter 5 in GH, but if you're interested:

1. look at Geometric Lorenz (§5.7)

Today:

1. Chapter 6: global bifurcations
   - Saddle-loop
   - Shil'nikov loop

2 Local bifurcations

So far we have been studying local bifurcations that happen near a fixed point or periodic orbit:

- Saddle-node
- Transcritical
- Pitchfork
- Hopf

which happens when a fixed point becomes non-hyperbolic (i.e. eigenvalues crossing the imaginary axis).

But! Even if all the fixed point are hyperbolic, we can still have bifurcations.

3 An example of global bifurcation

This is example 6.1.1 in GH. Suppose we have a system

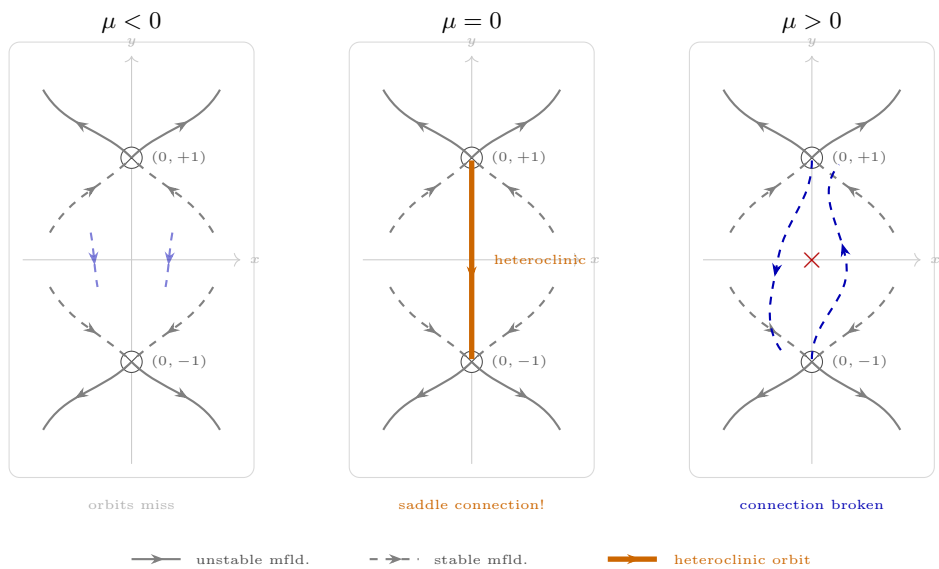
$$\dot{x} = \mu + x^2 - xy$$
$$\dot{y} = y^2 - x^2 - 1$$

If $\mu = 0$, then the fixed points are at

$$x(x - y) = 0$$
$$y^2 - x^2 - 1 = 0$$

and hence $(x_0, y_0) = (0, \pm 1)$.

Notice that at $\mu = 0$, we have a saddle connection between the two fixed points. But! When $\mu \neq 0$, this saddle connection breaks! Hence, the global behavior changes, but the fixed points are always hyperbolic.



This figure isn't entirely correct, the $\mu < 0$ case is not clear, but the $\mu = 0$ to $\mu > 0$ case captures the idea.

4 Example of saddle-loop global bifurcation

Suppose we have a system

$$\begin{aligned}\dot{x} &= y \\ \dot{y} &= x - x^2 + \mu y\end{aligned}$$

At $\mu = 0$, the system is Hamiltonian and we travel along level sets, getting a saddle loop. But! When $\mu \neq 0$, the saddle loop breaks and we loose all periodic orbits. Rowley has some plots of this that would be instructive to look at.

5 Local analysis near equilibrium

5.1 Setup

Consider the system

$$\begin{aligned}\dot{x} &= -\alpha x + f_1(x, y, \mu) \\ \dot{y} &= \beta y + f_2(x, y, \mu)\end{aligned}$$

for $\alpha, \beta > 0$ with f_1, f_2 being quadratic or higher in x, y. Let's also assume there is an equilibrium point at $(0, 0)$, causing the linear part to be

$$Df(0,0) = \begin{bmatrix} -\alpha & - \\ 0 & \beta \end{bmatrix}$$

Assume $\alpha \neq \beta$ do that the trace of $Df(0,0) \neq 0$. Also assume that for $\mu = 0$ we have a homoclinic orbit. This implies that there is some global bifurcation as μ crosses 0, but linearization tells us nothing. The main idea is to use local analysis near an equilibrium point to conclude some global behavior.

5.2 The technique

Let's look at trajectories that come near the equilibrium point. Near a fixed point, we can neglect the values of f_1, f_2 since they will be quite small.

The picture is quite graphical, but the idea is that we look at some trajectories near the fixed point in a box $[0, \epsilon] \times [0, \epsilon]$. We want to know how the trajectory intersects the top side wall $\Sigma_x = (\epsilon, y)$ and the top wall $\Sigma_y = (x, \epsilon)$.

Consider a Poincare map

$$P_0 : \Sigma_x \to \Sigma_y$$

on which we will perform linear analysis, and another Poincare map

$$P_1 : \Sigma_y \to \Sigma_x$$

Then we can define the system

$$P = P_1 \circ P_0 : \Sigma_x \to \Sigma_x$$

5.2.1 Analyzing the first map

Let's look at

$$P_0 : \Sigma_x \to \Sigma_y$$

starting at $(x_0, y_0) = (\epsilon, y_0)$. The solution of the linearized equations are

$$\begin{aligned} x(t) &= e^{-\alpha t} \\ y(t) &= e^{\beta t} y_0 \end{aligned}$$

Now what is the "time of flight"? Well, we leave the box when

$$\begin{aligned} y(T) &= \epsilon \\ \implies e^{\beta T} y_0 &= \epsilon \\ \beta T &= \log\left(\frac{\epsilon}{y_0}\right) \\ T &= \frac{1}{\beta} \log\left(\frac{\epsilon}{y_0}\right) \end{aligned}$$

Now where do we exit the box?

$$\begin{aligned} x(T) = e^{-\alpha T} &= e^{-\alpha \frac{1}{\beta} \log\left(\frac{\epsilon}{y_0}\right)} \\ &= \left(\frac{\epsilon}{y_0}\right)^{-\frac{\alpha}{\beta}} \epsilon \end{aligned}$$

Hence

$$P_0 : (\epsilon, y_0) \mapsto \left(\left(\frac{\epsilon}{y_0} \right)^{-\frac{\alpha}{\beta}} \epsilon, \epsilon \right)$$

5.2.2 Analyzing the second map

Let's look at

$$P_1 : \Sigma_y \to \Sigma_x$$

and Taylor expand about $x = 0, y = \epsilon, \mu = 0$, and keep the linear terms, to get

$$P_1(x, \epsilon, \mu) \approx (\epsilon, ax + b\mu)$$

What is the sign of a and b? I lost the plot of why we asked this, but hear is some argument. If $\mu = 0$, $x > 0 \implies y > 0$, so $a > 0$.

$$P_1(x, \epsilon) = (\epsilon, ax + \mu)$$

with rescaling μ, set $b = 1$.

5.2.3 Combine both maps

We combine both maps to write

$$P = P_1 \circ P_0 : (\epsilon, y) \mapsto P_1 \left(\left(\frac{\epsilon}{y_0} \right)^{-\frac{\alpha}{\beta}} \epsilon, \epsilon \right)$$

and hence

$$P(\epsilon, y_0) = \left(\epsilon, a \left(\frac{\epsilon}{y_0} \right)^{-\frac{\alpha}{\beta}} \epsilon + \mu \right)$$

Does P have fixed points? If so, the original system has a periodic orbit.

The Fixed point of P occurs at

$$y = a \left(\frac{\epsilon}{y_0} \right)^{-\frac{\alpha}{\beta}} \epsilon + \mu$$

If $\frac{\alpha}{\beta} > 1$, the there exist fixed points when $\mu > 0$. Hence, we have a periodic orbit for $\mu > 0$ and no periodic orbit for $\mu < 0$.

5.2.4 Stability of the Poincare map

We can write the stability of the Poincare map as

$$DP = \begin{pmatrix} 1 & 0 \\ 0 & \frac{\alpha}{\beta} A y^{\frac{\alpha}{\beta} - 1} \end{pmatrix}$$

where $\frac{\alpha}{\beta} A y^{\frac{\alpha}{\beta} - 1} \to 0$ as $y \to 0$. It is always $\frac{\alpha}{\beta} A y^{\frac{\alpha}{\beta} - 1} < 1$, which implies a stable periodic orbit. Note this is for $\alpha > \beta$ and $\mathrm{Tr}(Df) < 0$.

5.3 What happens for alpha less than beta

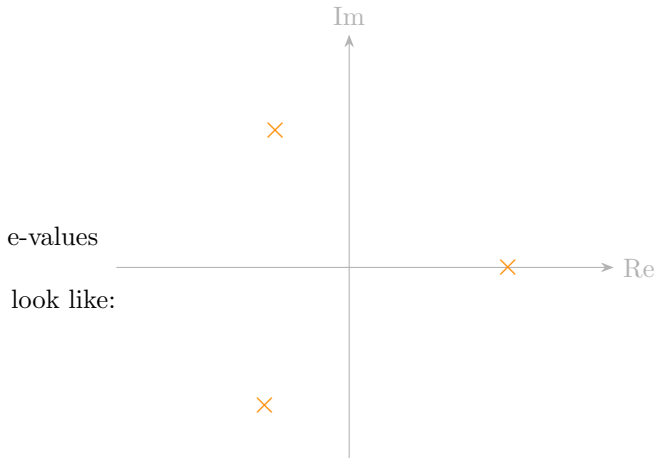
What happens when $\frac{\alpha}{\beta} < 1$? Well we have fixed points when $\mu \leq 0$. Hence there are periodic orbits when $\mu \leq 0$, but if we go through the math, these are unstable periodic orbits! Note this is for $\alpha < \beta$ and $\mathrm{Tr}(Df) > 0$.

6 Shil'nikov loop

Suppose we have the system

$$\dot{x} = f(x)$$

where $x \in \mathbb{R}^3$ and $x(0) = 0$ with eigenvalues that look like



e-values look like:

Also assume there is a homoclinic orbit.
Now Rowley is going through the an example numerically.

6.1 Theorem

The system we described above can be perturbed to a system that has infinitely many horseshoes.

6.1.1 Sketch of the theorem

Combine local analysis near fixed point with a global return map. Our local map looks like

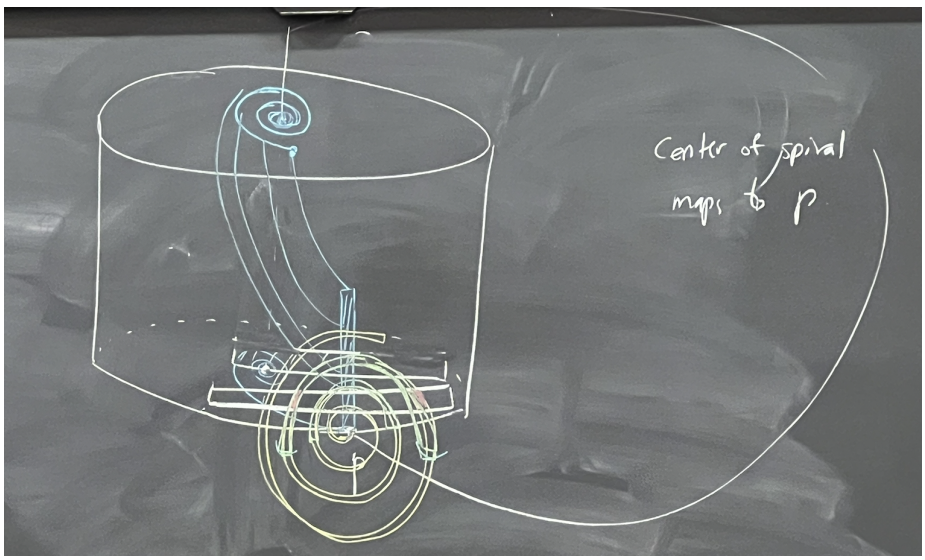
$$\begin{aligned}
\dot{x} &= \alpha x - \beta y + \cdots \\
\dot{y} &= \beta x + \alpha y + \cdots \\
\dot{z} &= \lambda z + \cdots
\end{aligned}$$

where α, β, λ correspond to the various coordinates of our eigenvalues, and in cylindrical coordinates we get

$$\begin{aligned}\dot{r} &= \alpha r + \cdots \\ \dot{\theta} &= \beta + \cdots \\ \dot{z} &= \lambda z + \cdots\end{aligned}$$

Now we look at a small cylinder about the fixed point. Define a mpa

$$P_0 : \text{ cylinder wall} \to \text{lid}$$

where we can get explicit solutions from our system in cylindrical coordinates.

The "time of flight" from the wall to the lid depends on λ and z_0. Okay, the explanation of how we get a Smale horseshoe from the Poincare maps is insane. Words are not sufficient.